COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
DATE PERFORMED:	09/19/26	**DATE SUBMITTED:**	09/19/26
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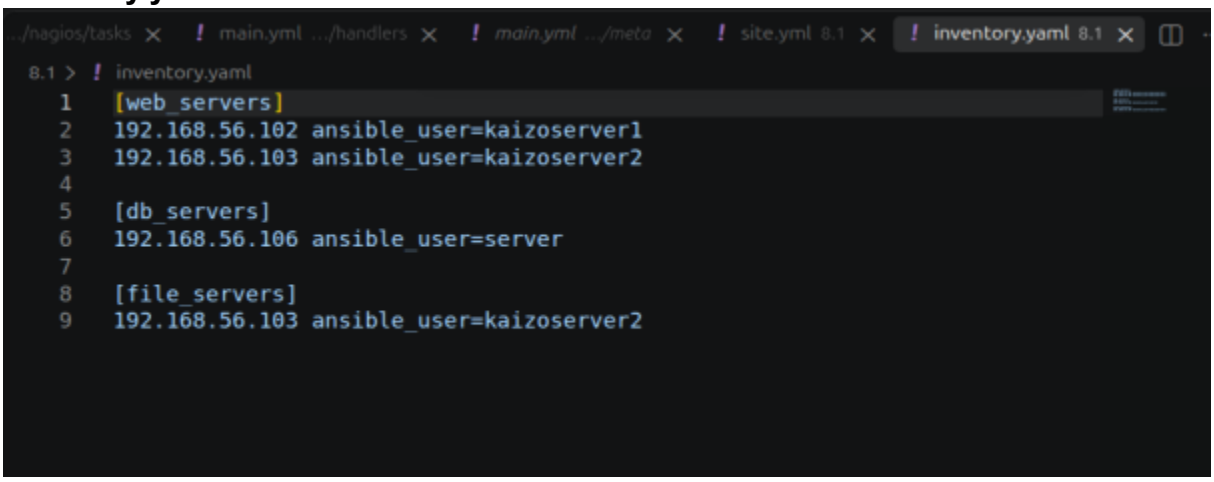
ACTIVITY NO. 8.1 **Install, Configure, and Manage Enterprise Availability Monitoring via Ansible**

1. PROCEDURE OUTPUT

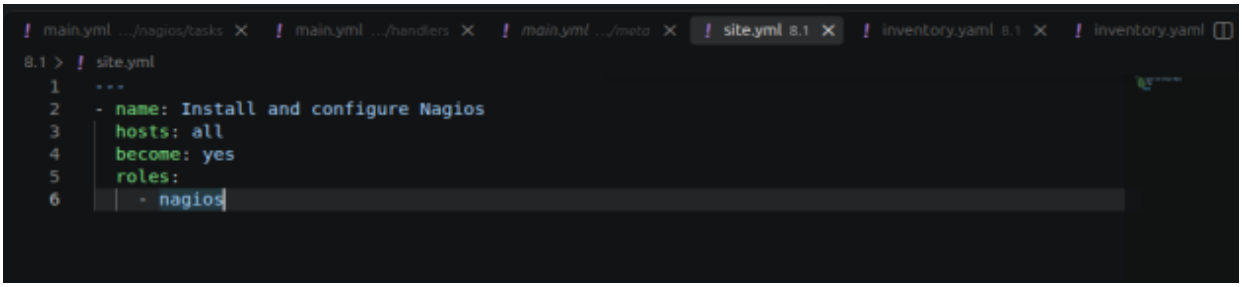
Creating roles for nagios

"ansible-galaxy init roles/nagios"(I forgot to screenshot this part and I accidentally clear the cmd)

inventory.yml



site.yml



roles/nagios/handlers/main.yml

```yaml
---
- name: restart nagios ubuntu
  ansible.builtin.service:
    name: nagios4
    state: restarted

- name: restart nagios centos
  ansible.builtin.service:
    name: nagios
    state: restarted
```

roles/nagios/tasks/main.yml

```yaml
---
- name: Install Nagios on Ubuntu
  ansible.builtin.apt:
    name:
      - nagios4
      - nagios-plugins-basic
    state: present
    update_cache: yes
  when: ansible_facts['os_family'] == "Debian"
  notify: restart nagios ubuntu

- name: Install EPEL repo on CentOS
  ansible.builtin.yum:
    name: epel-release
    state: present
  when: ansible_facts['os_family'] == "RedHat"

- name: Install Nagios on CentOS
  ansible.builtin.yum:
    name:
      - nagios
      - nagios-plugins-all
    state: present
  when: ansible_facts['os_family'] == "RedHat"
  notify: restart nagios centos

- name: Ensure Nagios is enabled and running
  ansible.builtin.service:
    name: "{{ 'nagios4' if ansible_facts['os_family'] == 'Debian' else 'nagios' }}"
    state: started
    enabled: yes
# tasks file for roles/nagios
```

RESULT:

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab/8.1$ ansible-playbook -b -K -i inventory.yaml site.yml
BECOME password:

PLAY [Install and configure Nagios] ********************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]
ok: [192.168.56.106]

TASK [nagios : Install Nagios on Ubuntu] ***************************************************************************
skipping: [192.168.56.106]
ok: [192.168.56.103]
ok: [192.168.56.102]

TASK [nagios : Install EPEL repo on CentOS] ************************************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]
ok: [192.168.56.106]

TASK [nagios : Install Nagios on CentOS] ***************************************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]
changed: [192.168.56.106]

TASK [nagios : Ensure Nagios is enabled and running] ***************************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]
changed: [192.168.56.106]

RUNNING HANDLER [nagios : restart nagios centos] *******************************************************************
changed: [192.168.56.106]

PLAY RECAP *********************************************************************************************************
192.168.56.102             : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.103             : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.106             : ok=5    changed=3    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

GIT:

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ git add .
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ git commit -m "HOA 8.1"
[main 917d5ac] HOA 8.1
 10 files changed, 158 insertions(+)
 create mode 100644 ansible-lab/8.1/inventory.yaml
 create mode 100644 ansible-lab/8.1/roles/nagios/README.md
 create mode 100644 ansible-lab/8.1/roles/nagios/defaults/main.yml
 create mode 100644 ansible-lab/8.1/roles/nagios/handlers/main.yml
 create mode 100644 ansible-lab/8.1/roles/nagios/meta/main.yml
 create mode 100644 ansible-lab/8.1/roles/nagios/tasks/main.yml
 create mode 100644 ansible-lab/8.1/roles/nagios/tests/inventory
 create mode 100644 ansible-lab/8.1/roles/nagios/tests/test.yml
 create mode 100644 ansible-lab/8.1/roles/nagios/vars/main.yml
 create mode 100644 ansible-lab/8.1/site.yml
```

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ git push origin main
Enumerating objects: 27, done.
Counting objects: 100% (26/26), done.
Delta compression using up to 3 threads
Compressing objects: 100% (14/14), done.
Writing objects: 100% (23/23), 3.27 KiB | 372.00 KiB/s, done.
Total 23 (delta 3), reused 4 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (3/3), completed with 2 local objects.
To https://github.com/ivnwynn/CPE212_SanJuan
   f1b0022..98fa9a8  main -> main
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What are the benefits of having an availability monitoring tool?

- Having an availability monitoring tool is important because it helps detect problems early, before they affect users or clients, which allows for a faster response and less downtime. It also keeps historical uptime data that can be useful for reporting and future capacity planning. Since alerts are automated, there's no need to constantly check servers manually, and having a centralized dashboard makes it easier to monitor multiple servers and services at once. Overall, it helps prevent revenue loss and reputational damage caused by unexpected downtime.

Conclusions/Learnings:

This activity taught me how to use Ansible roles to install and configure Nagios on both Ubuntu and CentOS. I ran into real issues along the way, an interrupted dpkg on Ubuntu, which I fixed with sudo dpkg --configure -a, and a dependency error on CentOS caused by missing Perl modules(the nagios-plugins-disk smb is not available in the repo of my Centos) , which I resolved by [enabling CRB / trimming the plugin list]. These problems helped me understand how differently package managers behave across distros. Overall, this showed me the value of Ansible for managing multiple servers with different OSes using a single playbook.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Claude**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code assistance, explanation of concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used AI to help me understand how to structure Ansible roles and to guide me**

through troubleshooting the errors I encountered, like the dpkg issue on Ubuntu and the dependency conflict on CentOS. I also used it to help me phrase parts of my report, like the conclusion and supplementary activity answers. However, I still ran all the commands myself, verified the outputs on my own VirtualBox setup, and made sure I understood why each fix worked before moving on. AI didn't do the activity for me — it helped me learn faster and fix problems I ran into along the way.

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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